11 Chemistry
Unit 2 Revision Worksheet

Details	
Student Name:	Teacher Name:
Date Set: ____/____/26	Due Date: ____/____/26

Curriculum Assessed

On completion of Unit 2 the student should be able to measure and analyse chemicals in water. This worksheet gives Easy, Medium and Hard practice on every Unit 2 topic: concentration, standard solutions and dilution, volumetric analysis, greenhouse gases, gases and pV = nRT, molar volume, limiting reactants, calorimetry and redox.

Task Overview

30 questions across 10 topics (87 marks): Easy 20 marks, Medium 27 marks, Hard 40 marks.
Answer in this booklet. Check your answers with the separate worked solutions.

Conditions

- Revision task (not assessed)
- VCE Chemistry Data Book
- Scientific calculator
- Suggested approach: do every Easy question first, then the Medium and Hard ones
- Aim for about 1.15 minutes per mark

Marking

- 87 marks
- Self-marked using the worked solutions.
- Show all working; give final answers with units and appropriate significant figures; include states in equations.

A+	A	B+	B	C+	C	D+	D	E+	E	NS
90-100%	80-89%	75-79%	70-74%	65-69%	60-64%	55-59%	50-54%	45-49%	40-44%	<40%

Total Marks:		Percentage:		Letter Grade:	

Topic 1 – Solution concentration and unit conversions

Each topic has one **Easy**, one **Medium** and one **Hard** question. Use the VCE Chemistry Data Book. Show all working, give final answers with units and appropriate significant figures, and include states in every equation. Worked solutions are in a separate file: try every question before you check.

Question 1 (2 marks) – Easy

A sports drink contains 30.0 g of sucrose, $C_{12}H_{22}O_{11}$, dissolved in every 500 mL of drink.

a) Calculate the concentration of sucrose in g L^{-1}.

1 mark

b) Calculate the concentration of sucrose in % (m/v).

1 mark

Question 2 (2 marks) – Medium

Epsom salts are magnesium sulfate, $MgSO_4$. A gardener's spray must contain 0.0400 M $MgSO_4$.

Calculate the mass of $MgSO_4$ the gardener must dissolve to make 250 mL of spray.

2 marks

Question 3 (4 marks) – Hard

A farmer controls algae in a dam by adding copper(II) sulfate, $CuSO_4$. A sample of the treated dam water is 0.0250% (m/v) $CuSO_4$.

a) Calculate the concentration of $CuSO_4$ in mol L^{-1}.

2 marks

b) Calculate the concentration of copper(II) ions, Cu^{2+}, in mg L^{-1}.

2 marks

Topic 2 – Standard solutions and dilution

Question 4 (1 mark) – Easy

A teacher needs 250.0 mL of 0.0640 M potassium iodide solution, KI(aq), and has a 0.800 M stock solution.

Calculate the volume of stock solution the teacher needs.

1 mark

Question 5 (3 marks) – Medium

Potassium hydrogen phthalate, $KHC_8H_4O_4$ (M = 204.2 g mol^{-1}), is a solid that can be weighed accurately, so it is used to make standard solutions.

a) Calculate the mass of $KHC_8H_4O_4$ needed to make 100.0 mL of a 0.0750 M standard solution.

2 marks

b) State one property of $KHC_8H_4O_4$ that makes it suitable as a primary standard.

1 mark

Question 6 (3 marks) – Hard

A 0.500 M sodium thiosulfate solution, $Na_2S_2O_3(aq)$, was diluted in two steps:

Step 1: 20.00 mL of the 0.500 M solution was diluted to 500.0 mL.

Step 2: 5.00 mL of the solution from Step 1 was diluted to 100.0 mL.

a) Calculate the concentration of $Na_2S_2O_3$ in the final solution, in mol L^{-1}.

2 marks

b) Convert your answer to part a) to mg L^{-1}.

1 mark

Topic 3 – Volumetric analysis

Question 7 (2 marks) – Easy

Before a titration the burette read 1.35 mL. At the end point it read 25.20 mL.

a) Calculate the titre.

1 mark

b) State the term used for the fixed volume of the solution being analysed that is placed in the conical flask.

1 mark

Question 8 (3 marks) – Medium

A 25.00 mL aliquot of limewater, calcium hydroxide solution, $Ca(OH)_2(aq)$, was titrated with 0.0800 M hydrochloric acid, HCl(aq). The average titre was 14.60 mL.

$$Ca(OH)_2(aq) + 2HCl(aq) \rightarrow CaCl_2(aq) + 2H_2O(l)$$

Calculate the concentration of calcium hydroxide in the limewater, in g L^{-1}.

3 marks

Question 9 (6 marks) – Hard

A soap maker uses a potassium hydroxide solution, KOH(aq), that is expected to be about 2.5 M. To check it, a student will titrate 25.00 mL aliquots against 0.100 M hydrochloric acid, HCl(aq). The burette holds 50.00 mL, and the student wants titres of about 20 mL.

$$KOH(aq) + HCl(aq) \rightarrow KCl(aq) + H_2O(l)$$

a) Calculate the concentration the KOH solution should have in each aliquot to give a titre of about 20 mL.

2 marks

b) Describe how the student should dilute the soap maker's solution, giving the glassware and volumes.

2 marks

c) Explain why titrating the undiluted KOH solution would not work with this equipment.

2 marks

Topic 4 – Greenhouse gases

Question 10 (2 marks) – Easy

For each of the greenhouse gases methane, CH_4, and carbon dioxide, CO_2, give one human activity that adds it to the atmosphere.

2 marks

Question 11 (2 marks) – Medium

Sunlight reaches Earth's surface mostly as visible light. The warmed surface then emits energy back towards space.

Explain, in terms of the radiation involved, how carbon dioxide molecules in the atmosphere cause the lower atmosphere to warm.

2 marks

Question 12 (4 marks) – Hard

The table shows the approximate concentrations of two gases measured at a clean-air monitoring station in 1980 and 2020. (ppm = parts per million)

Gas	1980 (ppm)	2020 (ppm)
carbon dioxide, CO_2	337	410
methane, CH_4	1.55	1.83

a) Calculate the percentage increase in the concentration of each gas between 1980 and 2020.

2 marks

b) Methane is present at a much lower concentration than carbon dioxide. Explain why scientists still consider the increase in methane important.

2 marks

Topic 5 – Gas pressure, SLC and pV = nRT

Question 13 (2 marks) – Easy

a) Convert 1.50 atm to kPa.

1 mark

b) Convert 45.0 °C to kelvin.

1 mark

Question 14 (3 marks) – Medium

A 3.00 L flask contains a gas at 100 kPa. The flask is connected to an empty (evacuated) 9.00 L flask and the tap between them is opened, so the gas fills both flasks. The temperature does not change.

a) Calculate the final pressure of the gas.

1 mark

b) Explain, in terms of the gas particles, why the pressure decreases.

2 marks

Question 15 (4 marks) – Hard

A weather balloon is filled at ground level with 30.0 mol of helium, He, at 101 kPa and 17.0 °C. Its volume at ground level is 716 L. The balloon rises to a height where the pressure is 30.0 kPa and the temperature is −40.0 °C.

a) Calculate the volume of the balloon at this height. Assume no helium escapes.

2 marks

b) The temperature falls as the balloon rises, which on its own would make the balloon shrink. Explain why the balloon still gets larger.

2 marks

Topic 6 – Molar volume and molar mass of gases

Question 16 (2 marks) – Easy

In an experiment at SLC, 0.0100 mol of a gas occupied 251 mL.

a) Calculate the experimental molar volume of the gas, in L mol^{-1}.

1 mark

b) Calculate the percentage difference between this result and the Data Book value at SLC, using the Data Book value as the reference.

1 mark

Question 17 (3 marks) – Medium

A 401 mL flask filled with an unknown gas has a mass of 0.712 g more than the empty (evacuated) flask. The gas is at 105 kPa and 40.0 °C.

a) Calculate the molar mass of the gas.

2 marks

b) Identify which of these gases it could be: methane, CH_4; oxygen, O_2; carbon dioxide, CO_2; argon, Ar.

1 mark

Question 18 (3 marks) – Hard

Yeast ferments glucose, $C_6H_{12}O_6$, in bread dough. The carbon dioxide, CO_2, produced makes the dough rise.

$$C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$$

Calculate the mass of glucose that must be fermented to produce 5.00 L of carbon dioxide, measured at SLC.

3 marks

Topic 7 – Limiting and excess reactants

Question 19 (2 marks) – Easy

Nitrogen, N_2, and hydrogen, H_2, react to form ammonia, NH_3:

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

2.0 mol of N_2 is mixed with 9.0 mol of H_2. Identify the limiting reactant and calculate the maximum amount, in mol, of NH_3 that can form.

2 marks

Question 20 (3 marks) – Medium

Copper, Cu, and sulfur, S, react when heated to form copper(I) sulfide, Cu_2S:

$$2Cu(s) + S(s) \rightarrow Cu_2S(s)$$

6.35 g of copper is heated with 1.28 g of sulfur. Identify the limiting reactant and calculate the mass of Cu_2S that forms.

3 marks

Question 21 (4 marks) – Hard

In a demonstration, zinc powder, Zn, reacts with iodine, I_2, to form zinc iodide, ZnI_2:

$$Zn(s) + I_2(s) \rightarrow ZnI_2(s)$$

6.54 g of zinc is mixed with 20.0 g of iodine and the reaction goes to completion.

a) Calculate the mass of zinc iodide formed.

3 marks

b) Calculate the mass of the excess reactant remaining.

1 mark

Topic 8 – Specific heat capacity and calorimetry

Question 22 (2 marks) – Easy

Olive oil has a specific heat capacity of 2.0 J g^{-1} $°C^{-1}$.

a) Calculate the energy, in kJ, needed to heat 1.20 kg of water, H_2O, from 15.0 °C to 65.0 °C.

1 mark

b) The same amount of energy is transferred to 1.20 kg of olive oil. Determine whether the temperature rise of the oil would be greater or smaller than that of the water, and by what factor.

1 mark

Question 23 (3 marks) – Medium

An instant cold pack works because dissolving ammonium nitrate, NH_4NO_3, absorbs energy. A student dissolved 8.00 g of NH_4NO_3 in 100.0 g of water, H_2O, in an insulated cup. The temperature fell from 22.0 °C to 16.1 °C. Assume all the energy came from the water.

a) Calculate the energy absorbed when the NH_4NO_3 dissolved, in J.

1 mark

b) Calculate the energy absorbed per mole of NH_4NO_3, in kJ mol^{-1}.

2 marks

Question 24 (5 marks) – Hard

200 g of water, H_2O, at 80.0 °C is poured into 300 g of water at 20.0 °C in a polystyrene cup.

a) Calculate the final temperature of the mixture, assuming no energy is lost.

3 marks

b) The measured final temperature was 41.0 °C. Calculate the energy, in kJ, that was not accounted for by the cold water.

2 marks

Topic 9 – Redox: oxidising agents, reducing agents and equations

Question 25 (2 marks) – Easy

Sodium, Na, burns in chlorine gas, Cl_2:

$$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$$

a) Identify the species that is oxidised.

1 mark

b) Identify the oxidising agent.

1 mark

Question 26 (2 marks) – Medium

Roadside breath-testing devices once used orange dichromate ions, $Cr_2O_7^{2-}$, which are reduced to green chromium(III) ions, Cr^{3+}, in acidic conditions.

Write the balanced reduction half-equation, including states.

2 marks

Question 27 (3 marks) – Hard

Hydrogen peroxide, H_2O_2, oxidises iodide ions, I^-, to iodine, I_2, in acidic conditions. The hydrogen peroxide is reduced to water.

a) Write the oxidation half-equation, including states.

1 mark

b) Write the reduction half-equation, including states.

1 mark

c) Write the balanced overall equation, including states.

1 mark

Topic 10 – Reactivity series, corrosion and primary cells

Question 28 (3 marks) – Easy

A coil of copper wire, Cu, is placed in colourless silver nitrate solution, $AgNO_3(aq)$, and left for an hour.

a) State one observation that shows a reaction has occurred.

1 mark

b) Write the balanced overall ionic equation, including states.

2 marks

Question 29 (3 marks) – Medium

A silver oxide, Ag_2O, button cell, used in watches, is a simple primary cell. Its half-equations are:

$$Zn(s) + 2OH^-(aq) \rightarrow ZnO(s) + H_2O(l) + 2e^-$$

$$Ag_2O(s) + H_2O(l) + 2e^- \rightarrow 2Ag(s) + 2OH^-(aq)$$

a) Identify the oxidising agent in the cell.

1 mark

b) Write the overall equation for the cell reaction.

1 mark

c) Over its life, the cell transfers 0.0100 mol of electrons. Calculate the mass of zinc used up.

1 mark

Question 30 (4 marks) – Hard

A wad of damp iron wool, Fe, was pushed into the bottom of a measuring cylinder. The cylinder was turned upside down in a dish of water, trapping 100.0 mL of air at SLC. Over a week, the iron wool rusted and the water rose inside the cylinder until the trapped gas had a volume of 79.5 mL at SLC.

The overall equation for the rusting is $4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$.

a) Identify the oxidising agent in the rusting reaction.

1 mark

b) Calculate the mass of iron that rusted.

3 marks

Additional space

Clearly number any response written here.

END OF WORKSHEET